EX 8	Reading and Writing on External Storage (SD card) with Persistent and Time-based Notifications

Aim

To implement file reading and writing on external storage (SD card) and to use Notification Manager to display persistent and time-based notifications in Android.

Definitions

External Storage

External storage is a storage location on an Android device that allows applications to store and access files outside the app's private internal storage, and it can be accessed by the user and other apps.

Reading and Writing on External Storage

Reading and writing on external storage is the process of storing data into a file and retrieving data from a file located in the device's external storage such as an SD card.

Notification Manager

Notification Manager is a system service in Android that is used to create, display, update, and cancel notifications in the device's notification area.

Persistent Notification

Persistent notification is a notification that stays in the notification bar and cannot be cleared by the user, indicating that an application or service is running continuously.

Time-based Notification

A time-based notification is a notification that is triggered and displayed after a specific time interval or at a scheduled time to notify the user.

Procedure

1. Open Android Studio IDE → File → New → New Project → specify the application name "SD Card" and company domain "com.mad.sdcard" → click "next" → choose Minimum SDK "API 17:Android 4.2(Jelly Bean)" → click "Next" → choose "Blank Activity" → click "next" → specify the Activity Name "MainActivity" → click "Finish".
2. Open MainActivity.java under app/java/ sdcard.mad.com.sdcard and type the following codes:

MainActivity.java

```
package sdcard.mad.com.sdcard;

import android.app.Activity;
import android.app.Notification;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.os.Environment;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import java.io.BufferedReader;
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.InputStreamReader;
import android.support.v4.app.NotificationCompat;

public class MainActivity extends Activity {

    EditText editText;
    TextView textView;

    Button writeBtn, readBtn, notifyBtn, timerBtn, clearBtn;

    String filename = "sample.txt";

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editText = (EditText) findViewById(R.id.editText);
```

```
textView = (TextView) findViewById(R.id.textView);

writeBtn = (Button) findViewById(R.id.btnWrite);
readBtn = (Button) findViewById(R.id.btnRead);
notifyBtn = (Button) findViewById(R.id.btnNotify);
timerBtn = (Button) findViewById(R.id.btnTimer);
clearBtn = (Button) findViewById(R.id.btnClear);   // CLEAR BUTTON INITIALIZED

// WRITE FILE TO SD CARD
writeBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        try {

            File path = Environment.getExternalStorageDirectory();
            File file = new File(path, filename);

            FileOutputStream fos = new FileOutputStream(file);
            fos.write(editText.getText().toString().getBytes());
            fos.close();

            Toast.makeText(MainActivity.this,
                    "File Written to SD Card",
                    Toast.LENGTH_LONG).show();

        } catch (Exception e) {
            e.printStackTrace();
        }

    }
});

// READ FILE FROM SD CARD
readBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        try {

            File path = Environment.getExternalStorageDirectory();
            File file = new File(path, filename);

            FileInputStream fis = new FileInputStream(file);

            BufferedReader br = new BufferedReader(
                    new InputStreamReader(fis));

            String line;
            String data = "";
```

```
            while ((line = br.readLine()) != null) {
                data = data + line;
            }

            br.close();

            textView.setText(data);

            Toast.makeText(MainActivity.this,
                    "File Read Successful",
                    Toast.LENGTH_LONG).show();

        } catch (Exception e) {
            e.printStackTrace();
        }

    }
});

// CLEAR TEXT BUTTON
clearBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        editText.setText("");
        textView.setText("");

        Toast.makeText(MainActivity.this,
                "Text Cleared",
                Toast.LENGTH_SHORT).show();
    }
});

// PERSISTENT NOTIFICATION
notifyBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        NotificationManager nm =
                (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

        Intent intent = new Intent(MainActivity.this, MainActivity.class);

        PendingIntent pi = PendingIntent.getActivity(
                MainActivity.this, 0, intent, 0);

        Notification notification = new NotificationCompat.Builder(MainActivity.this)
                .setSmallIcon(R.mipmap.ic_launcher)
                .setContentTitle("Application Running")
```

```
                .setContentText("This notification cannot be cleared")
                .setTicker("Persistent Notification")
                .setContentIntent(pi)
                .setOngoing(true)
                .build();

            nm.notify(1, notification);

        }
    });

    // TIME BASED NOTIFICATION (5 seconds delay)
    timerBtn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            Handler handler = new Handler();

            handler.postDelayed(new Runnable() {
                @Override
                public void run() {

                    NotificationManager nm =
                        (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

                    Intent intent = new Intent(MainActivity.this, MainActivity.class);

                    PendingIntent pi = PendingIntent.getActivity(
                        MainActivity.this, 0, intent, 0);

                    Notification notification = new
NotificationCompat.Builder(MainActivity.this)
                        .setSmallIcon(R.mipmap.ic_launcher)
                        .setContentTitle("Reminder")
                        .setContentText("Notification after 5 seconds")
                        .setTicker("Timer Notification")
                        .setContentIntent(pi)
                        .build();

                    nm.notify(2, notification);

                }
            }, 5000);

        }
    });

  }

}
```

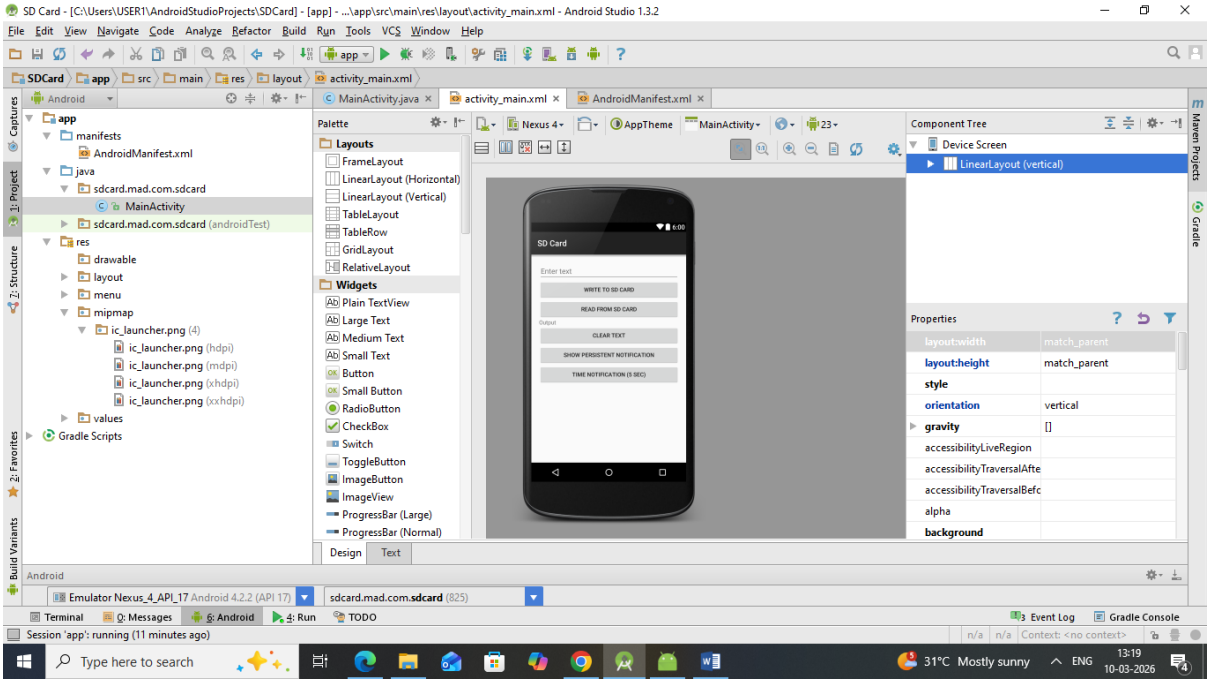
3. Open activity_main.xml under app/res/layout and type the following codes:

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/editText"
        android:hint="Enter text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnWrite"
        android:text="Write to SD Card"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnRead"
        android:text="Read from SD Card"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/textView"
        android:text="Output"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnClear"
        android:text="Clear Text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnNotify"
        android:text="Show Persistent Notification"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnTimer"
        android:text="Time Notification (5 sec)"
        android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"/>
</LinearLayout>
```

4. The design of the application will be as follows:



5. Open AndroidManifest.xml under app/manifests folder and do the following changes:

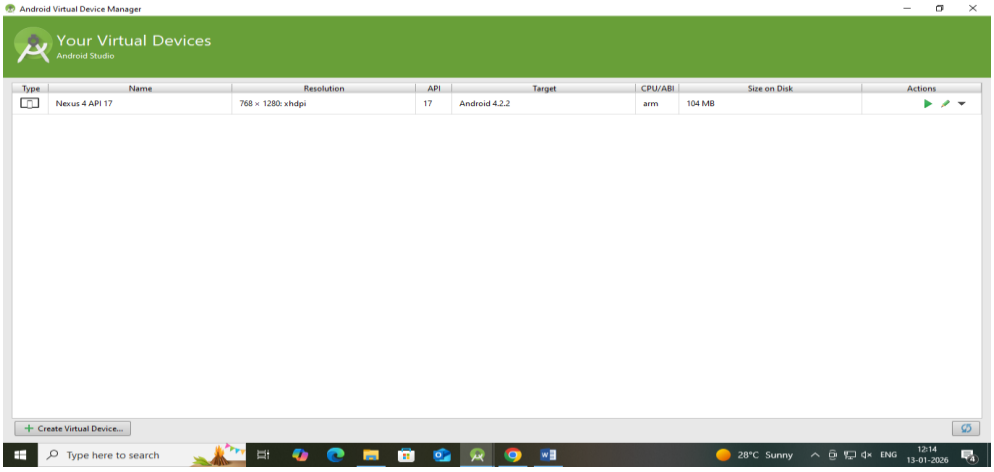
AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   package="sdcard.mad.com.sdcard" >

   <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
   <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>

   <application
      android:allowBackup="true"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:theme="@style/AppTheme" >
      <activity
         android:name=".MainActivity"
         android:label="@string/app_name" >
         <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER" />
         </intent-filter>
```

```
    </activity>
  </application>

</manifest>
```

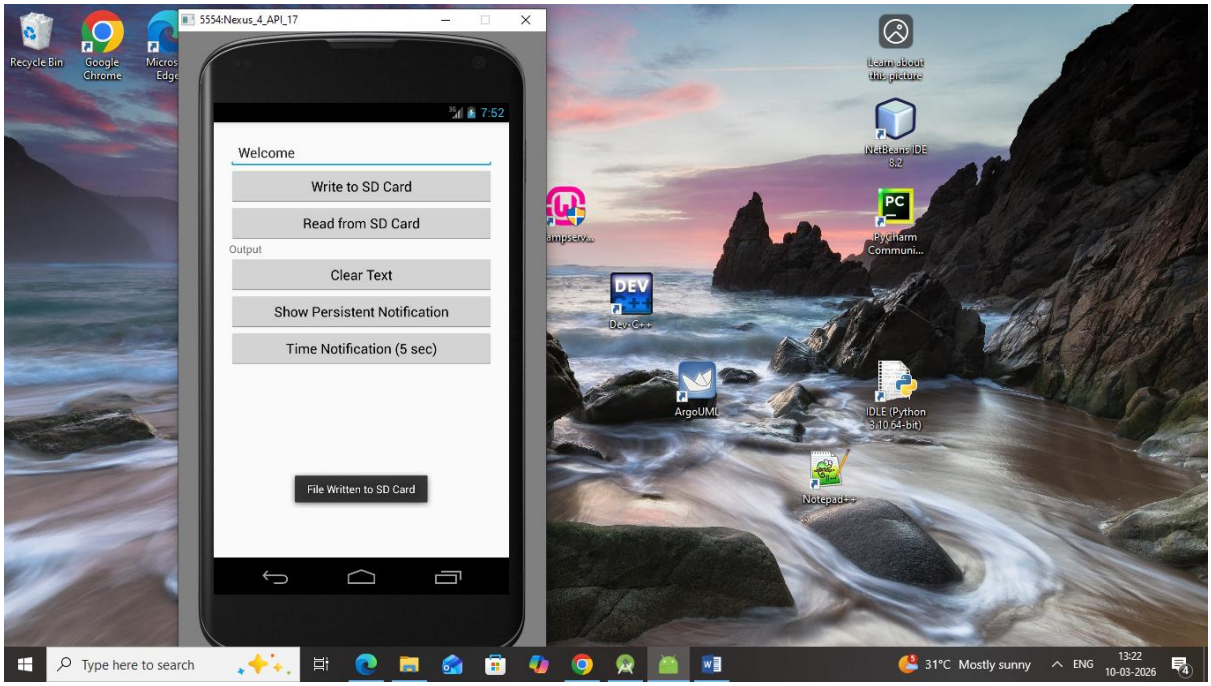
6. Go to Tools → android → AVD Manager → click "+ create a virtual device" → select "phone" from category → select "Nexus 4" from the list → click "next" → select Release name: Jelly Bean, API Level: 17, ABI: armeabi-v7a, Target: Android 4.2.2 from the list → click "next" → Choose orientation "portrait" → click "finish".

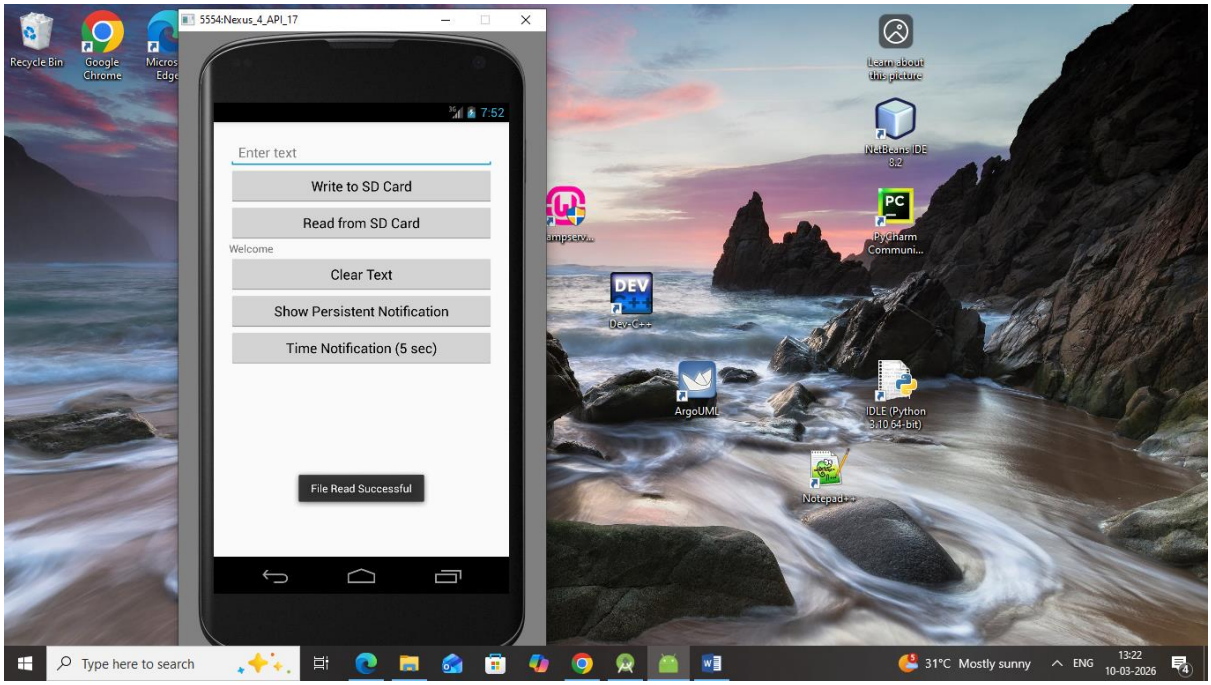
The following window will appear after configuring AVD:

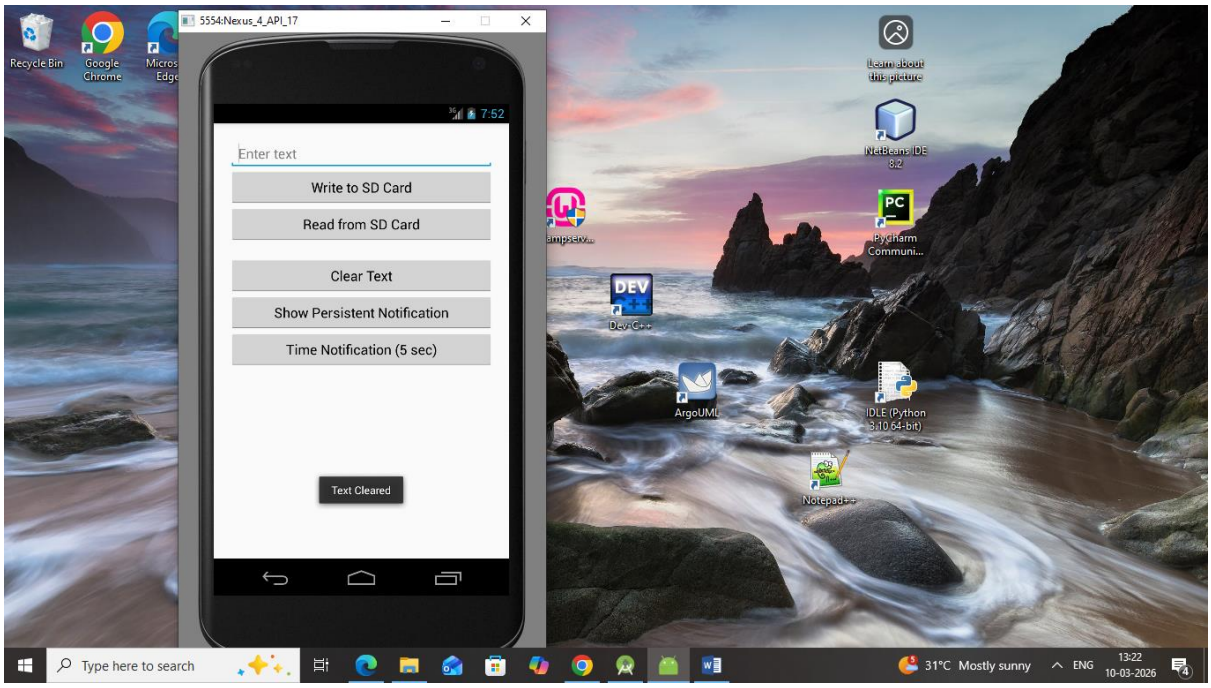


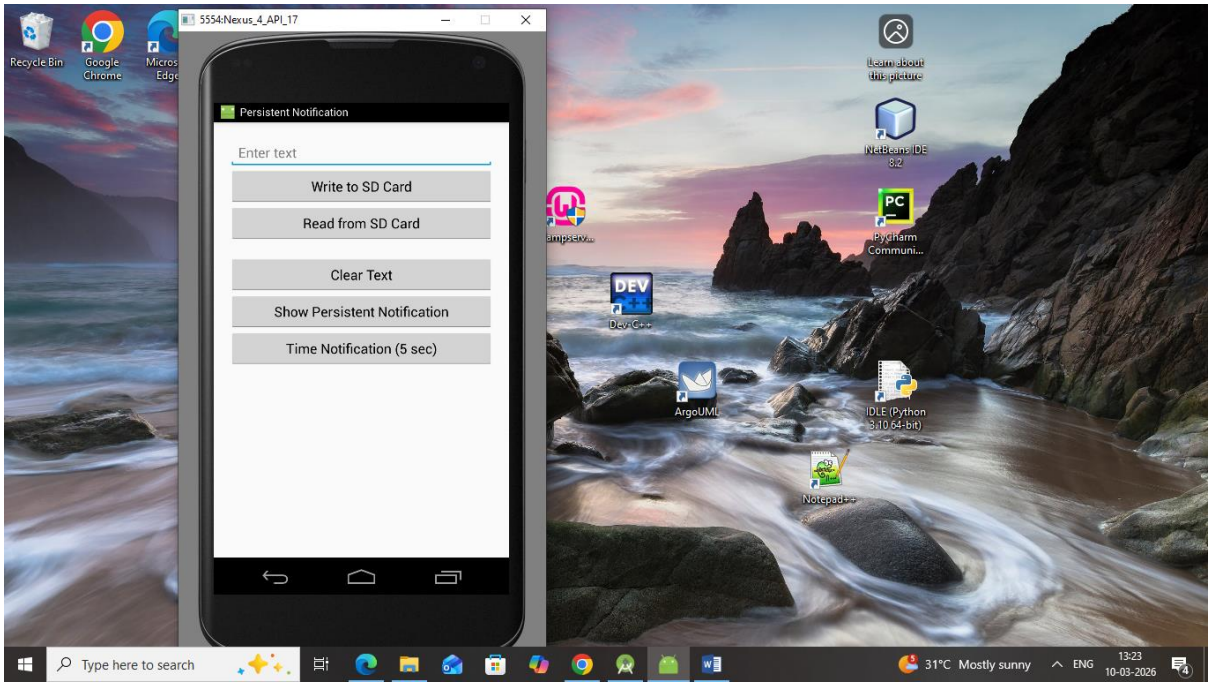
7. Click "Run app" button in the Android Studio → choose android virtual device → click "ok".

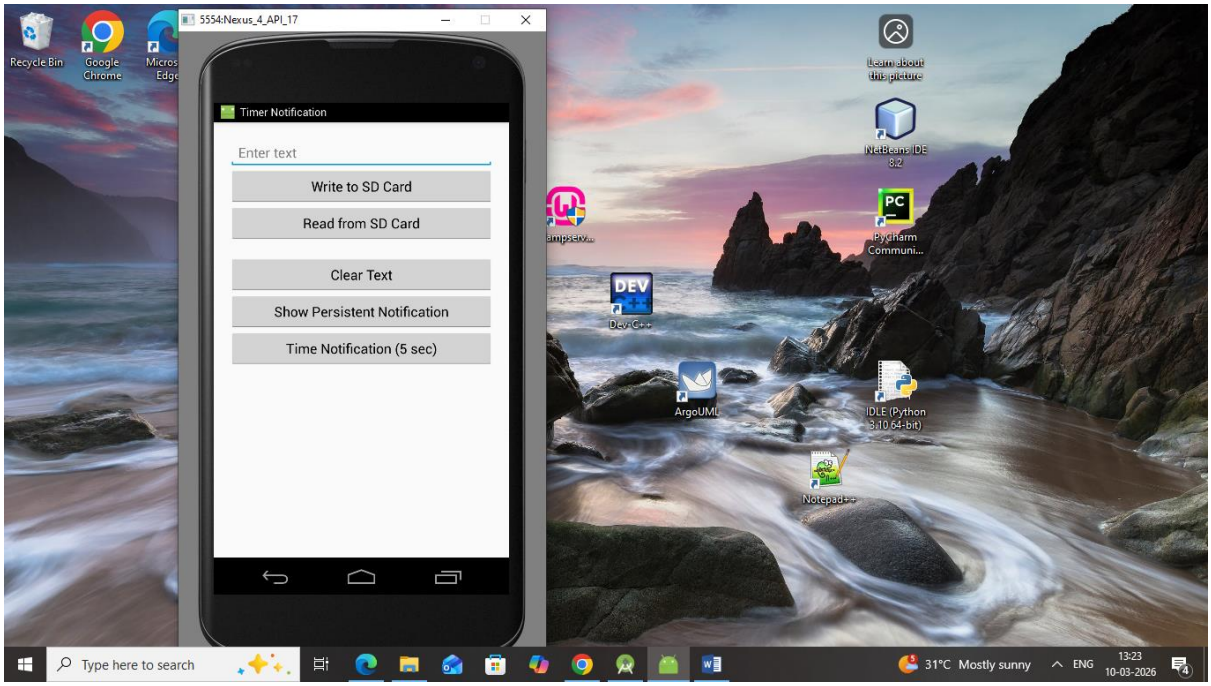
Output

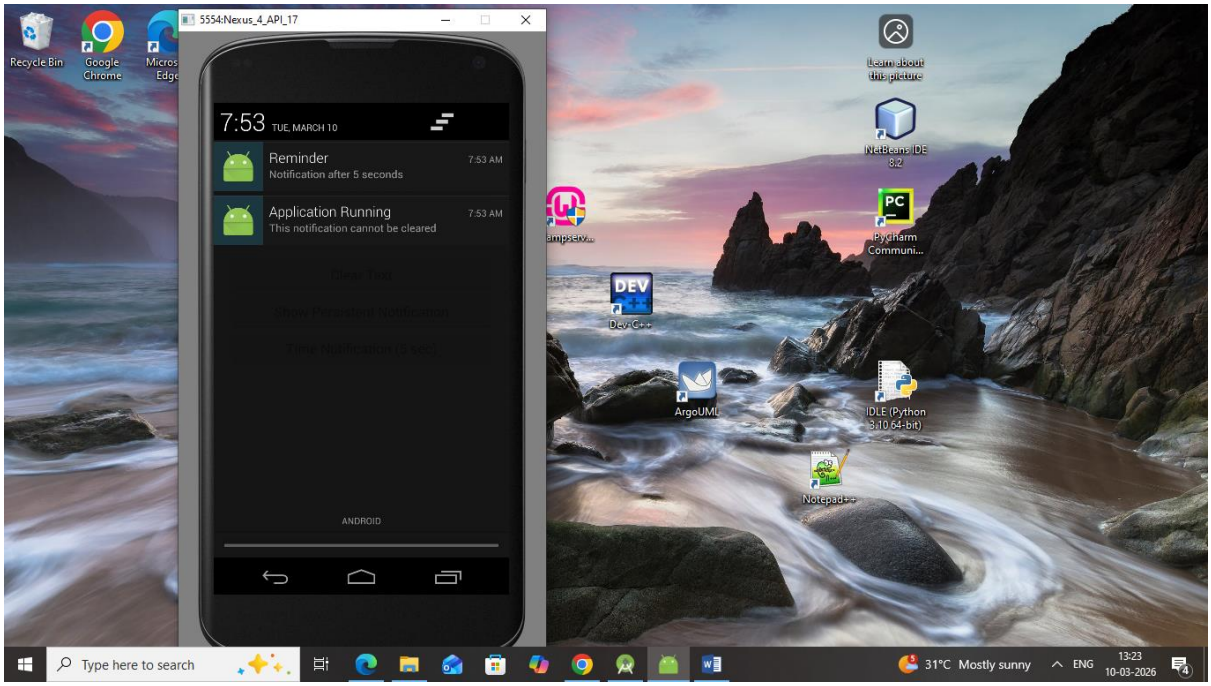



5554:Nexus_4_API_17
7:52
Enter text
Write to SD Card
Read from SD Card
Welcome
Clear Text
Show Persistent Notification
Time Notification (5 sec)
File Read Successful


5554:Nexus_4_API_17
7:52
Enter text
Write to SD Card
Read from SD Card
Clear Text
Show Persistent Notification
Time Notification (5 sec)
Text Cleared


5554:Nexus_4_API_17
Persistent Notification
Enter text
Write to SD Card
Read from SD Card
Clear Text
Show Persistent Notification
Time Notification (5 sec)


5554:Nexus_4_API_17
Timer Notification
Enter text
Write to SD Card
Read from SD Card
Clear Text
Show Persistent Notification
Time Notification (5 sec)


5554:Nexus_4_API_17
7:53 TUE, MARCH 10
Reminder
Notification after 5 seconds
7:53 AM
Application Running
This notification cannot be cleared
7:53 AM
ANDROID
Recycle Bin
Google Chrome
Learn about this picture
NetBeans IDE 8.2
PyCharm Communi...
Dev-C++
ArgoUML
IDLE (Python 3.10 64-bit)
Notepad++
Type here to search
31°C Mostly sunny
ENG
13-23
10-03-2026

Result

Thus, file reading and writing on external storage (SD card) with Notification Manager to display persistent and time-based notifications have been implemented in Android.